```cpp

// ----------SEMESTER 02 FINAL OOP PROJECT-------------
// ---------TOPIC: UNIVERSITY LOGIN PORTAL-------------


#include <iostream>
#include <string>
#include <vector>
#include <fstream>
#include <iomanip>
#include <cstdlib>
using namespace std;

// GLOBAL DATA
// simple vectors
vector<string> studentData;
vector<string> teacherData;
vector<string> courseData;
vector<string> marksData;

// HELPER FUNCTION
//
string getWord(string s, int pos){
    int count = 1;
    string word = "";
    for(int i=0; i<s.length(); i++){
        if(s[i]==',')
            count++;
        else if(count==pos)
            word += s[i];
    }
    return word;
}

// Convert string to int
int toInt(string s){
    int num = 0;
    for(int i=0; i<s.length(); i++){
        num = num*10 + (s[i]-'0');
    }
    return num;
}

//  FILE HANDLING

void saveToFile(){
    ofstream fout; // file out

    fout.open("students.txt");
    for(int i=0; i<studentData.size(); i++){
        fout<<studentData[i]<<endl;
    }
    fout.close();

    fout.open("teachers.txt");
```

```
    for(int i=0; i<teacherData.size(); i++){
        fout<<teacherData[i]<<endl;
    }
    fout.close();

    fout.open("courses.txt");
    for(int i=0; i<courseData.size(); i++){
        fout<<courseData[i]<<endl;
    }
    fout.close();

    fout.open("marks.txt");
    for(int i=0; i<marksData.size(); i++){
        fout<<marksData[i]<<endl;
    }
    fout.close();

    cout<<"All Data Saved Successfully!"<<endl;
}

// Load data from files when program starts
void loadFromFile(){
    string line;
    ifstream fin; // file in

    fin.open("students.txt");
    while(getline(fin, line)){
        if(line!="") studentData.push_back(line);
    }
    fin.close();

    fin.open("teachers.txt");
    while(getline(fin, line)){
        if(line!="") teacherData.push_back(line);
    }
    fin.close();

    fin.open("courses.txt");
    while(getline(fin, line)){
        if(line!="") courseData.push_back(line);
    }
    fin.close();

    fin.open("marks.txt");
    while(getline(fin, line)){
        if(line!="") marksData.push_back(line);
    }
    fin.close();
}

//  GPA CALCULATOR
double getGPA(int marks){
    if(marks >= 90) return 4.0;
    else if(marks >= 85) return 3.7;
    else if(marks >= 80) return 3.5;
```

```
111:     else if(marks >= 75) return 3.0;
112:     else if(marks >= 70) return 2.7;
113:     else if(marks >= 65) return 2.5;
114:     else if(marks >= 60) return 2.0;
115:     else if(marks >= 55) return 1.7;
116:     else if(marks >= 50) return 1.0;
117:     else return 0.0;
118: }
119:
120: string getGrade(int marks){
121:     if(marks >= 90) return "A";
122:     else if(marks >= 80) return "B";
123:     else if(marks >= 70) return "C";
124:     else if(marks >= 60) return "D";
125:     else return "F";
126: }
127:
128: //  ADMIN FUNCTIONS
129: void adminAddStudent(){
130:     int id; string name, user, pass;
131:     cout<<"\nEnter Student ID: "; cin>>id;
132:     cout<<"Enter Student Name: "; cin>>name;
133:     cout<<"Enter Username: "; cin>>user;
134:     cout<<"Enter Password: "; cin>>pass;
135:     string data = to_string(id)+","+name+","+user+","+pass;
136:     studentData.push_back(data);
137:     cout<<"Student Added Successfully!"<<endl;
138: }
139:
140: void adminViewStudent(){
141:     system("cls");
142:     cout<<"\n========== STUDENT LIST ==========="<<endl;
143:     cout<<left<<setw(10)<<"ID"<<setw(20)<<"NAME"<<setw(15)<<"USERNAME"<<endl;
144:     cout<<"-----------------------------------"<<endl;
145:     for(int i=0; i<studentData.size(); i++){
146:         cout<<getWord(studentData[i],1)<<"\t"<<getWord(studentData[i],2)<<"\t"<<getWord(stude
147:     }
148: }
149:
150: void adminAddTeacher(){
151:     int id; string name, user, pass, dept;
152:     cout<<"\nEnter Teacher ID: "; cin>>id;
153:     cout<<"Enter Teacher Name: "; cin>>name;
154:     cout<<"Enter Username: "; cin>>user;
155:     cout<<"Enter Password: "; cin>>pass;
156:     cout<<"Enter Department: "; cin>>dept;
157:     string data = to_string(id)+","+name+","+user+","+pass+","+dept;
158:     teacherData.push_back(data);
159:     cout<<"Teacher Added Successfully!"<<endl;
160: }
161:
162: void adminViewTeacher(){
163:     system("cls");
164:     cout<<"\n========== TEACHER LIST ==========="<<endl;
165:     cout<<left<<setw(10)<<"ID"<<setw(20)<<"NAME"<<setw(15)<<"DEPT"<<endl;
```

```
166:     cout<<"-----------------------------------"<<endl;
167:     for(int i=0; i<teacherData.size(); i++){
168:         cout<<getWord(teacherData[i],1)<<"\t"<<getWord(teacherData[i],2)<<"\t"<<getWord(teach
169:     }
170: }
171:
172: void adminAddCourse(){
173:     int id, tid, credit;
174:     string name;
175:     cout<<"\nEnter Course ID: "; cin>>id;
176:     cout<<"Enter Course Name: "; cin>>name;
177:     cout<<"Enter Teacher ID: "; cin>>tid;
178:     cout<<"Enter Credit Hours: "; cin>>credit;
179:     string data = to_string(id)+","+name+","+to_string(tid)+","+to_string(credit);
180:     courseData.push_back(data);
181:     cout<<"Course Added Successfully!"<<endl;
182: }
183:
184: void adminViewCourse(){
185:     system("cls");
186:     cout<<"\n========== COURSE LIST ==========="<<endl;
187:     cout<<left<<setw(10)<<"ID"<<setw(20)<<"NAME"<<setw(10)<<"TID"<<setw(10)<<"CREDIT"<<endl;
188:     cout<<"-----------------------------------"<<endl;
189:     for(int i=0; i<courseData.size(); i++){
190:         cout<<getWord(courseData[i],1)<<"\t"<<getWord(courseData[i],2)<<"\t"<<getWord(courseD
191:     }
192: }
193:
194: void adminDeleteStudent(){
195:     int id; cout<<"Enter Student ID to Delete: "; cin>>id;
196:     for(int i=0; i<studentData.size(); i++){
197:         if(toInt(getWord(studentData[i],1))==id){
198:             studentData.erase(studentData.begin()+i);
199:             cout<<"Student Deleted!"<<endl;
200:             return;
201:         }
202:     }
203:     cout<<"Student Not Found!"<<endl;
204: }
205:
206: //  TEACHER FUNCTIONS
207: void teacherViewAllStudents(){
208:     system("cls");
209:     cout<<"\n========== ALL STUDENTS ==========="<<endl;
210:     for(int i=0; i<studentData.size(); i++){
211:         cout<<"ID: "<<getWord(studentData[i],1)<<" Name: "<<getWord(studentData[i],2)<<endl;
212:     }
213: }
214:
215: void teacherEnterMarks(){
216:     int sid, cid, marks;
217:     cout<<"\nEnter Student ID: "; cin>>sid;
218:     cout<<"Enter Course ID: "; cin>>cid;
219:     cout<<"Enter Marks out of 100: "; cin>>marks;
220:     string data = to_string(sid)+","+to_string(cid)+","+to_string(marks);
```

```
221:     marksData.push_back(data);
222:     cout<<"Marks Saved!"<<endl;
223: }
224:
225: void teacherViewMarks(){
226:     system("cls");
227:     cout<<"\n========== ALL MARKS ==========="<<endl;
228:     cout<<left<<setw(10)<<"SID"<<setw(10)<<"CID"<<setw(10)<<"MARKS"<<setw(10)<<"GRADE"<<endl;
229:     for(int i=0; i<marksData.size(); i++){
230:         int m = toInt(getWord(marksData[i],3));
231:         cout<<getWord(marksData[i],1)<<"\t"<<getWord(marksData[i],2)<<"\t"<<m<<"\t"<<getGrade
232:     }
233: }
234:
235: void teacherSearchStudent(){
236:     int id; cout<<"Enter Student ID to Search: "; cin>>id;
237:     for(int i=0; i<studentData.size(); i++){
238:         if(toInt(getWord(studentData[i],1))==id){
239:             cout<<"Found! Name: "<<getWord(studentData[i],2)<<endl;
240:             return;
241:         }
242:     }
243:     cout<<"Not Found!"<<endl;
244: }
245:
246: // STUDENT FUNCTIONS
247: void studentViewProfile(int sid){
248:     system("cls");
249:     for(int i=0; i<studentData.size(); i++){
250:         if(toInt(getWord(studentData[i],1))==sid){
251:             cout<<"\n========== YOUR PROFILE ==========="<<endl;
252:             cout<<"ID: "<<getWord(studentData[i],1)<<endl;
253:             cout<<"Name: "<<getWord(studentData[i],2)<<endl;
254:             cout<<"Username: "<<getWord(studentData[i],3)<<endl;
255:             return;
256:         }
257:     }
258: }
259:
260: void studentViewCourses(){
261:     system("cls");
262:     cout<<"\n========== AVAILABLE COURSES ==========="<<endl;
263:     cout<<left<<setw(10)<<"ID"<<setw(20)<<"NAME"<<setw(10)<<"CREDITS"<<endl;
264:     for(int i=0; i<courseData.size(); i++){
265:         cout<<getWord(courseData[i],1)<<"\t"<<getWord(courseData[i],2)<<"\t"<<getWord(courseD
266:     }
267: }
268:
269: void studentViewResult(int sid){
270:     system("cls");
271:     double totalGPA=0; int count=0; int totalMarks=0;
272:     cout<<"\n========== YOUR RESULT ==========="<<endl;
273:     cout<<left<<setw(10)<<"CID"<<setw(10)<<"MARKS"<<setw(10)<<"GPA"<<setw(10)<<"GRADE"<<endl;
274:     for(int i=0; i<marksData.size(); i++){
275:         if(toInt(getWord(marksData[i],1))==sid){
```

```
276:             int marks = toInt(getWord(marksData[i],3));
277:             double gpa = getGPA(marks);
278:             cout<<getWord(marksData[i],2)<<"\t"<<marks<<"\t"<<gpa<<"\t"<<getGrade(marks)<<end
279:             totalGPA += gpa; count++; totalMarks += marks;
280:         }
281:     }
282:     if(count>0){
283:         cout<<"-----------------------------------"<<endl;
284:         cout<<"Total Subjects: "<<count<<endl;
285:         cout<<"Total Marks: "<<totalMarks<<endl;
286:         cout<<"Your CGPA: "<<totalGPA/count<<endl;
287:     }else{
288:         cout<<"No result found yet!"<<endl;
289:     }
290: }
291:
292: //  LOGIN CHECK
293: bool checkPassword(){
294:     string pass;
295:     cout<<"Enter Password: ";
296:     cin>>pass;
297:     if(pass=="123"){
298:         cout<<"Login Success!"<<endl;
299:         system("pause");
300:         return true;
301:     }else{
302:         cout<<"Wrong Password!"<<endl;
303:         system("pause");
304:         return false;
305:     }
306: }
307:
308: //  MAIN MENU
309: int main(){
310:     loadFromFile(); // load old data
311:
312:     // Add default data if file is empty
313:     if(studentData.size()==0){
314:         studentData.push_back("1,Ali,ali,123");
315:         studentData.push_back("2,Sara,sara,123");
316:     }
317:     if(teacherData.size()==0){
318:         teacherData.push_back("101,SirAhmad,ahmad,123,CS");
319:         teacherData.push_back("102,MissSara,sara,123,Math");
320:     }
321:     if(courseData.size()==0){
322:         courseData.push_back("201,OOP,101,3");
323:         courseData.push_back("202,Math,102,3");
324:     }
325:
326:     int mainChoice;
327:     while(1){
328:         system("cls");
329:         cout<<"\n====================================="<<endl;
330:         cout<<" WELCOME TO UNISPHERE SYSTEM "<<endl;
```

```
331:         cout<<"====================================="<<endl;
332:         cout<<"1. Admin Panel"<<endl;
333:         cout<<"2. Teacher Panel"<<endl;
334:         cout<<"3. Student Panel"<<endl;
335:         cout<<"4. Save and Exit"<<endl;
336:         cout<<"====================================="<<endl;
337:         cout<<"Enter Your Choice: ";
338:         cin>>mainChoice;
339:
340:         if(mainChoice==1){ // ADMIN
341:             if(checkPassword()==false) continue;
342:             int adminChoice;
343:             while(1){
344:                 system("cls");
345:                 cout<<"\n========== ADMIN MENU ==========="<<endl;
346:                 cout<<"1. Add New Student"<<endl;
347:                 cout<<"2. View All Students"<<endl;
348:                 cout<<"3. Delete Student"<<endl;
349:                 cout<<"4. Add New Teacher"<<endl;
350:                 cout<<"5. View All Teachers"<<endl;
351:                 cout<<"6. Add New Course"<<endl;
352:                 cout<<"7. View All Courses"<<endl;
353:                 cout<<"8. Save Data"<<endl;
354:                 cout<<"9. Logout"<<endl;
355:                 cout<<"Enter Choice: ";
356:                 cin>>adminChoice;
357:
358:                 if(adminChoice==1) adminAddStudent();
359:                 else if(adminChoice==2) adminViewStudent();
360:                 else if(adminChoice==3) adminDeleteStudent();
361:                 else if(adminChoice==4) adminAddTeacher();
362:                 else if(adminChoice==5) adminViewTeacher();
363:                 else if(adminChoice==6) adminAddCourse();
364:                 else if(adminChoice==7) adminViewCourse();
365:                 else if(adminChoice==8) saveToFile();
366:                 else if(adminChoice==9) break;
367:                 else cout<<"Wrong Choice!"<<endl;
368:                 system("pause");
369:             }
370:         }
371:
372:         else if(mainChoice==2){ // TEACHER
373:             if(checkPassword()==false) continue;
374:             int teacherChoice;
375:             while(1){
376:                 system("cls");
377:                 cout<<"\n========== TEACHER MENU ==========="<<endl;
378:                 cout<<"1. View All Students"<<endl;
379:                 cout<<"2. Search Student"<<endl;
380:                 cout<<"3. Enter Student Marks"<<endl;
381:                 cout<<"4. View All Marks"<<endl;
382:                 cout<<"5. Logout"<<endl;
383:                 cout<<"Enter Choice: ";
384:                 cin>>teacherChoice;
385:
```

```
386:                 if(teacherChoice==1) teacherViewAllStudents();
387:                 else if(teacherChoice==2) teacherSearchStudent();
388:                 else if(teacherChoice==3) teacherEnterMarks();
389:                 else if(teacherChoice==4) teacherViewMarks();
390:                 else if(teacherChoice==5) break;
391:                 else cout<<"Wrong Choice!"<<endl;
392:                 system("pause");
393:             }
394:         }
395:
396:         else if(mainChoice==3){ // STUDENT
397:             if(checkPassword()==false) continue;
398:             int sid;
399:             cout<<"Enter Your Student ID: ";
400:             cin>>sid;
401:
402:             int studentChoice;
403:             while(1){
404:                 system("cls");
405:                 cout<<"\n========== STUDENT MENU ==========="<<endl;
406:                 cout<<"1. View My Profile"<<endl;
407:                 cout<<"2. View All Courses"<<endl;
408:                 cout<<"3. View My Result and GPA"<<endl;
409:                 cout<<"4. Logout"<<endl;
410:                 cout<<"Enter Choice: ";
411:                 cin>>studentChoice;
412:
413:                 if(studentChoice==1) studentViewProfile(sid);
414:                 else if(studentChoice==2) studentViewCourses();
415:                 else if(studentChoice==3) studentViewResult(sid);
416:                 else if(studentChoice==4) break;
417:                 else cout<<"Wrong Choice!"<<endl;
418:                 system("pause");
419:             }
420:         }
421:
422:         else if(mainChoice==4){
423:             saveToFile();
424:             cout<<"\nThank You For Using UniSphere!"<<endl;
425:             break;
426:         }
427:         else{
428:             cout<<"Invalid Choice!"<<endl;
429:             system("pause");
430:         }
431:     }
432:
433:     return 0;
434: }
```